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DEPARTMENT OF COMPUTER ENGINEERING

CSL5605 Cloud Computing Lab

Sixth Semester, 2023-2024 (Even Semester)

Name of Student : **Mainuddin Nayeem Khan**

Roll No. : **53**

Division : **A**

Batch :**I**

Day / Session : **Monday**

Venue : **307**

Experiment No. :**2**

Title of Experiment : To study and implement Hosted Virtualization using Virtual Box& KVM.

Date of Conduction : **19-2-26**

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment 2

Aim:

To study and implement Hosted Virtualization using Virtual Box& KVM.

Theory:

1. Explain virtualization and types of Virtualization.

Virtualization is a technology that allows a single physical computer system to be divided into multiple virtual environments. Each virtual environment behaves like an independent system with its own operating system, applications, and resources, even though all of them share the same physical hardware. Virtualization is achieved using software called a hypervisor, which manages and allocates hardware resources such as CPU, memory, storage, and network to the virtual machines.

Types of Virtualization

1. Server Virtualization**:-**
   In server virtualization, a physical server is divided into multiple virtual servers. Each virtual server runs its own operating system and applications. This helps in better utilization of hardware, reduces cost, and improves scalibility.

2. Storage Virtualization**:-**
   Storage virtualization combines multiple physical storage devices into a single virtual storage unit. It simplifies storage management, improves backup and recovery, and provides better flexibility in data handling.

3. Network Virtualization**:-**
   Network virtualization creates virtual networks by combining hardware and software network resources. It allows multiple virtual networks to run on the same physical network infrastructure, improving efficiency and security.

4. Desktop Virtualization**:-**
   Desktop virtualization allows user desktops to run on a centralized server instead of individual machines. Users can access their desktops remotely from any device, which improves data security and ease of management.

5. Application Virtualization**:-**
   Application virtualization enables applications to run in an isolated environment without being installed directly on the user’s operating system. This reduces compatibility issues and makes application deployment easier.

2. Explain KVM.

KVM (Kernel-based Virtual Machine) is a Linux-based virtualization solution that is freely available and open source.
It enables the Linux kernel to directly manage virtual machines by

acting as a bare-metal (Type-1) hypervisor, allowing several virtual systems to operate on a single physical machine.

Key Features of KVM

- Utilizes CPU virtualization technologies such as Intel VT-x and AMD-V.
- Ensures efficient performance with strong scalability.
- Allows different operating systems to run as guest machines.
- Commonly implemented in enterprise and cloud. environments.

3. What is a Virtual Box?

VirtualBox is a Type-2 (hosted) hypervisor created by Oracle that enables a single computer to run multiple operating systems at the same time.

Features of VirtualBox

- Simple installation with an easy-to-use interface.
- Compatible with Windows, Linux, and macOS platforms.
- Commonly used for education, software testing, and development tasks.

4. Role of Virtualization in cloud computing.

Virtualization forms the foundation of cloud computing by enabling the abstraction of physical hardware.

It allows cloud providers to deliver computing resources dynamically based on user demand.

Role of Virtualization

- Optimizes the utilization of physical resources.
- Simplifies the creation and control of virtual machines.
- Supports scalability and operational flexibility.
- Minimizes hardware investment and infrastructure expenses.

5. Advantage and limitation of virtualization.

Advantages of Virtualization

- Better utilization of hardware resources.
- Reduces cost of physical infrastructure.
- Improves scalability and flexibility.
- Enables easy backup, recovery, and management.

Limitations of Virtualization

- Initial setup and configuration can be complex.
- Performance overhead compared to physical systems.
- Requires skilled administrators.
- Hardware compatibility issues may arise.

Installation of Operating System on Virtual Machine

1. Explain steps to install OS on VM using KVM/Virtual Box with screenshots.

Steps to Install an Operating System on a Virtual Machine using VirtualBox

Step 1: Install VirtualBox Software

1. Download Oracle VirtualBox from its official website.
2. Execute the setup file and complete the installation process.
3. Open VirtualBox after successful installation.

Step 2: Download the OS ISO Image

1. Obtain the ISO file of the required operating system such as Ubuntu, Linux Mint, or Windows.
2. Store the ISO file on your computer for later use.

Step 3: Create a Virtual Machine

1. Launch VirtualBox and click on the New option.
2. Provide a suitable name for the virtual machine.
3. Choose the operating system type and version.
4. Proceed by clicking Next.

Step 4: Assign Memory (RAM)

1. Allocate the required RAM to the virtual machine.
2. A minimum of 2 GB RAM is recommended for proper performance.
3. Click Next.

Step 5: Configure Virtual Storage

1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image) as the disk type.
3. Select Dynamically allocated storage.
4. Specify disk size (at least 20 GB recommended).
5. Click Create.

Step 6: Load the OS ISO File

1. Select the virtual machine and open Settings.
2. Navigate to Storage and choose Empty under Controller IDE.
3. Click the disk icon and browse to select the OS ISO file.
4. Confirm by clicking OK.

Step 7: Start the Virtual Machine

1. Select the configured virtual machine.
2. Click Start to boot the system.
3. The installer will load from the attached ISO file.

Step 8: Install the Operating System

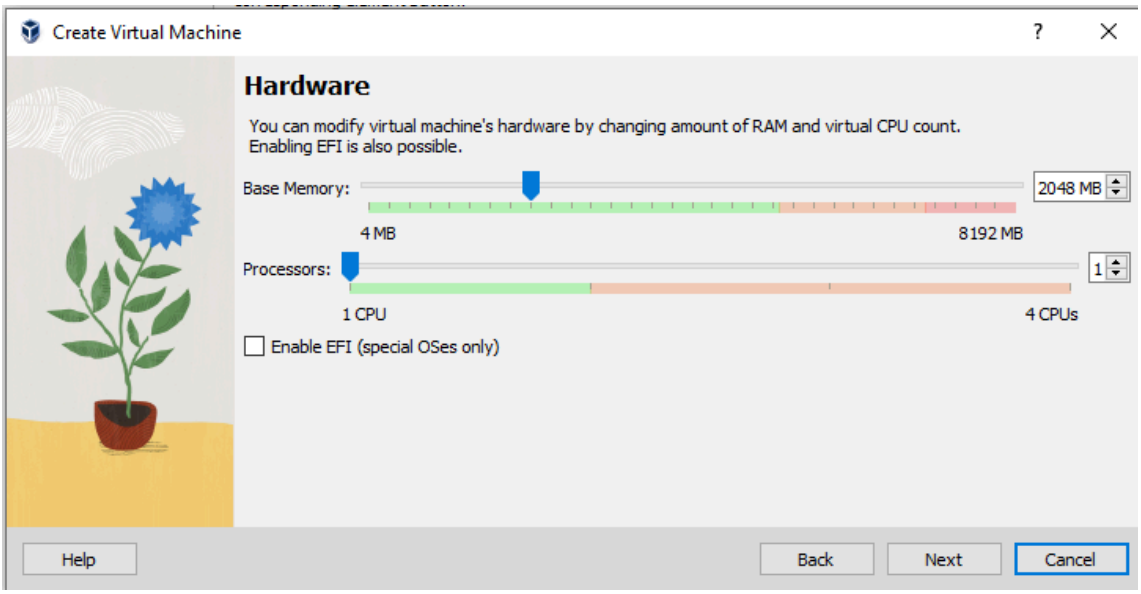
1. Choose language and keyboard preferences.
2. Select the install option and complete user setup details.
3. Follow on-screen instructions until installation finishes.

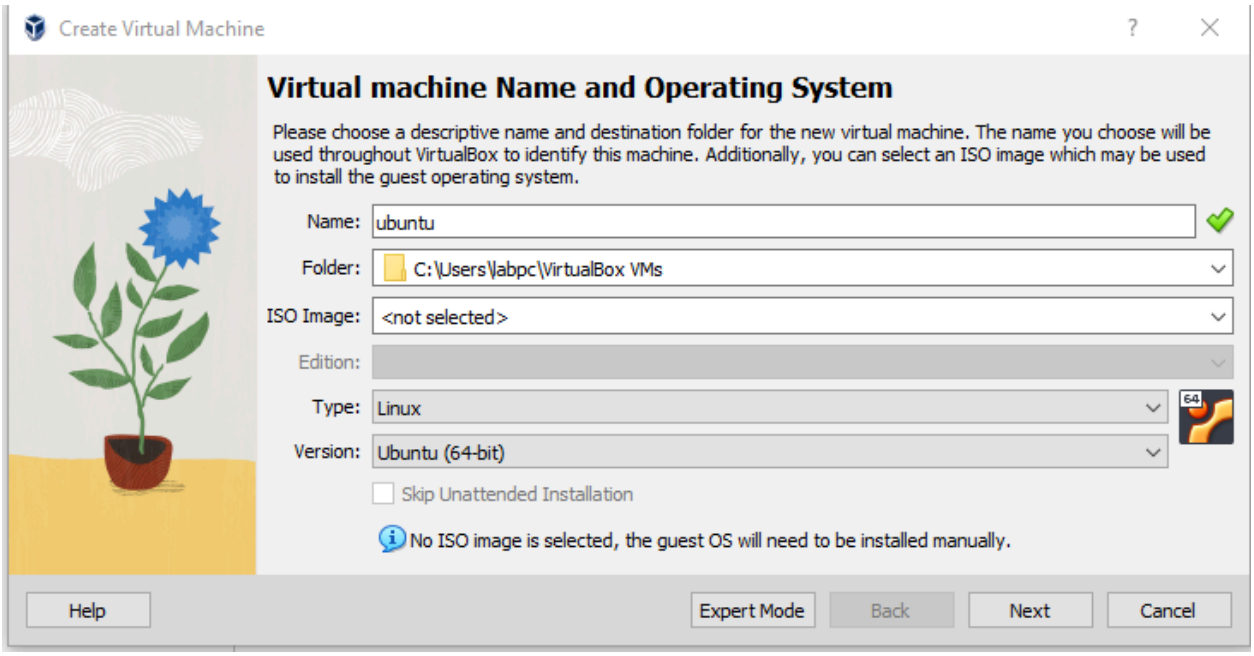
Step 9: Restart the System

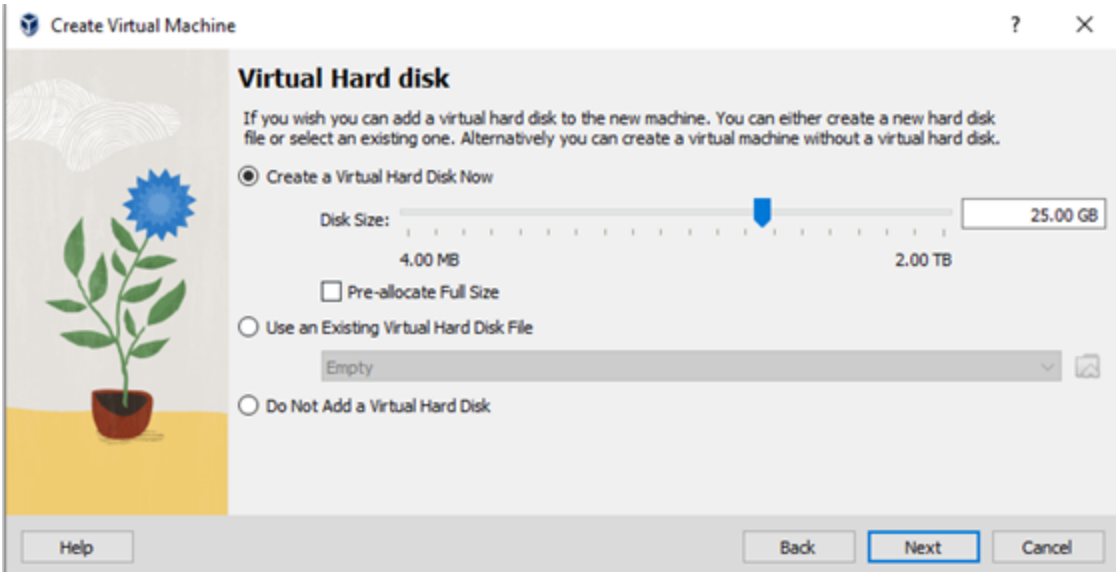
1. Restart the virtual machine after installation.
2. Remove the ISO file when prompted.
3. The VM will boot into the newly installed operating system.

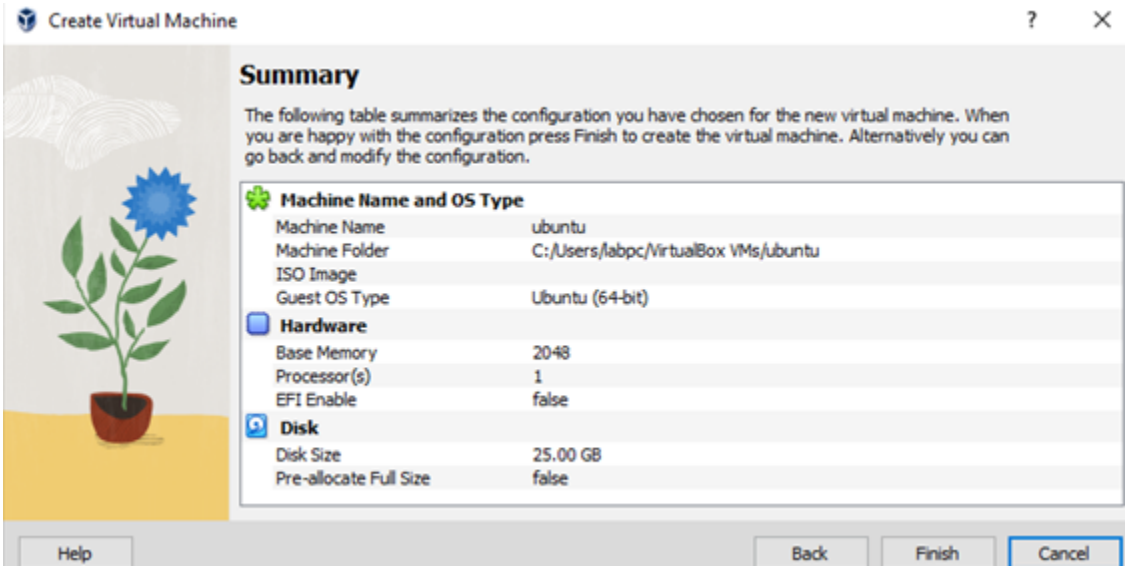
Step 10: Install Guest Additions (Optional)

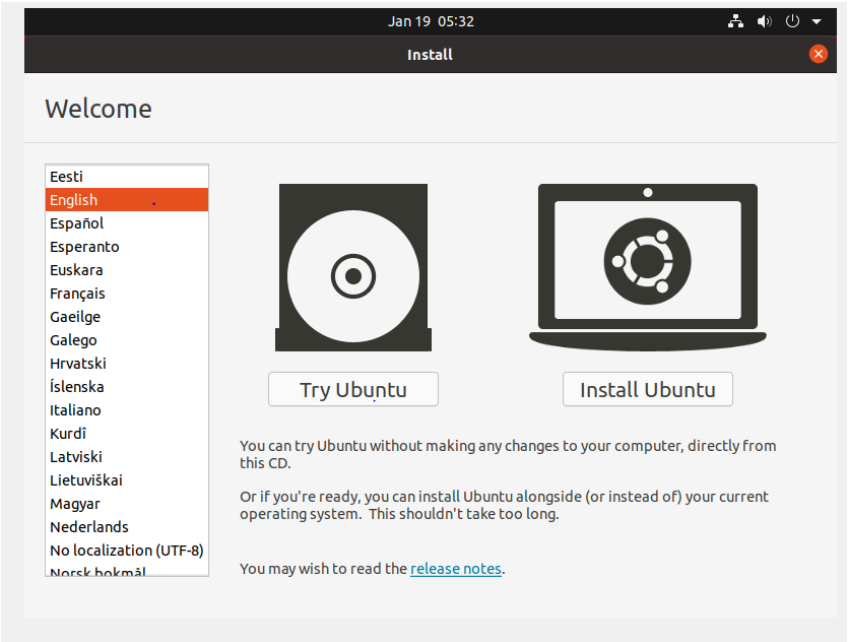
1. From the menu, select Devices → Insert Guest Additions CD Image.
2. Install Guest Additions inside the virtual machine.
3. Restart the VM to enhance performance and display features.

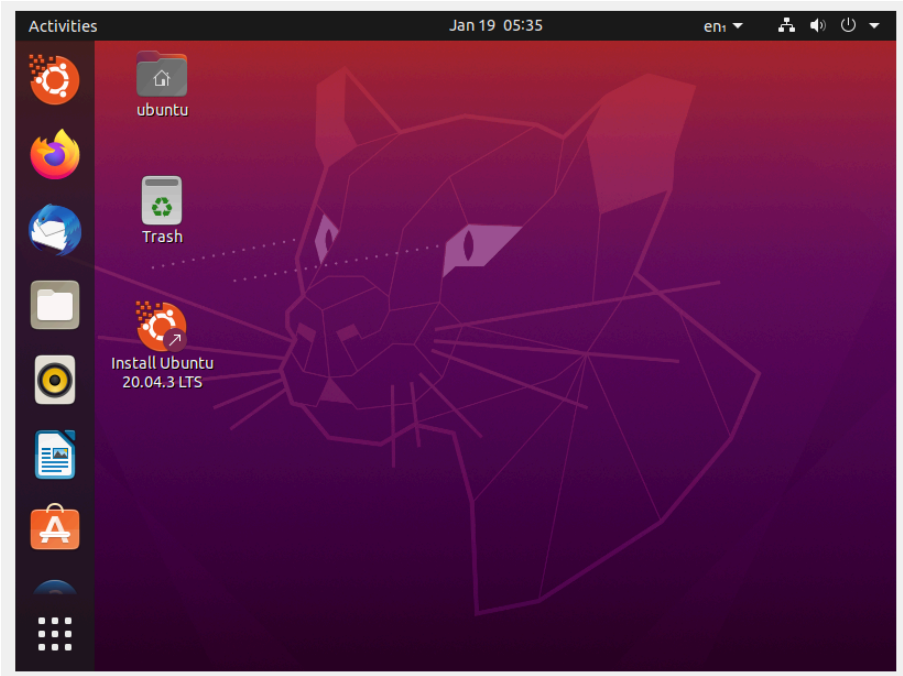





Create Virtual Machine
Virtual Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.
Create a Virtual Hard Disk Now
Disk Size:
25.00 GB
4.00 MB
2.00 TB
Pre-allocate Full Size
Use an Existing Virtual Hard Disk File
Empty
Do Not Add a Virtual Hard Disk
Help
Back
Next
Cancel


Create Virtual Machine
Summary
The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.
Machine Name and OS Type
Machine Name ubuntu
Machine Folder C:/Users/labpc/VirtualBox VMs/ubuntu
ISO Image
Guest OS Type Ubuntu (64-bit)
Hardware
Base Memory 2048
Processor(s) 1
EFI Enable false
Disk
Disk Size 25.00 GB
Pre-allocate Full Size false
Help
Back
Finish
Cancel


Jan 19 05:32
Install
Welcome
Eesti
English
Español
Esperanto
Euskara
Français
Gaeilge
Galego
Hrvatski
Íslenska
Italiano
Kurdî
Latviski
Lietuviškai
Magyar
Nederlands
No localization (UTF-8)
Try Ubuntu
Install Ubuntu
You can try Ubuntu without making any changes to your computer, directly from this CD.
Or if you're ready, you can install Ubuntu alongside (or instead of) your current operating system. This shouldn't take too long.
You may wish to read the release notes.


Activities
Jan 19 05:35
en
ubuntu
Trash
Install Ubuntu
20.04.3 LTS

CONCLUSION

Virtualization enables multiple operating systems to run on a single physical

machine, improving resource utilization and reducing cost. It is a core technology

in cloud computing and provides scalability, flexibility, and efficient system management.

1) Criteria to Select a Specific Type of Hypervisor:

- Type-1 hypervisors are preferred for servers and cloud environments
- Type-2 hypervisors are suitable for desktops and learning
- Performance and scalability requirements
- Hardware compatibility and virtualization support
- Security and cost considerations

2) How to Resolve Limitations of Virtualization:

- Use hardware-assisted virtualization
- Properly allocate CPU, memory, and storage
- Keep hypervisor and VMs updated
- Monitor resources to avoid overload
- Use strong security and management tools